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12 R

Reg. No. | S | A | N | T | H | O | S | H |

Time : 3.00 hrs.

Quarterly Examination - 2026

Max. Marks : 70

PHYSICS

PART - I

Choose the correct option: **15x1=15**

1. Two identical conducting balls having positive charges q_1 and q_2 are separated by a distance r. If they are made to touch each other and then separated to the same distance, now the force between them will be a) less than before b) same as before c) more than before d) zero
2. If three 2μF capacitors are connected in parallel, then the effective capacitance is
   a) 2 μF b) 4 μF c) 6 μF d) 3 μF
3. In an equipotential surface, a change of 5C is moved between two points. The work done is
   a) 5 J b) 2 J c) zero d) 10 J
4. A toaster operating at 240V has a resistance of 120Ω It's power is a) 400W b) 2W c) 480W d) 240W
5. The internal resistance of a 2.1V Cell, which gives a current of 0.2A through a resistance of 10Ω is
   a) 0.2 Ω b) 0.5 Ω c) 0.8 Ω d) 1.0 Ω
6. In a Joules heating law, when R and t are constant, If H is taken along the Y-axis and I^2 along x-axis, the graph is a) straight line b) parabola c) circle d) ellipse
7. A particle having mass m and charge q accelerated through a potential difference V. Find the force experienced when it is kept under perpendicular magnetic field $\vec{B}$.
   a) $\sqrt{\frac{2q^3BV}{m}}$ b) $\sqrt{\frac{q^3B^2V}{2m}}$ c) $\sqrt{\frac{2q^3B^2V}{m}}$ d) zero
8. A wire of length of l carrying a current i along the y-direction is kept in a magnetic field given by $\vec{B} = \frac{\beta}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ T. The magnitude of Lorentz force acting on the wire is
   a) $\sqrt{\frac{2}{3}}$ BIℓ b) $\sqrt{\frac{1}{3}}$ BIℓ c) $\sqrt{2}$ BIℓ d) $\sqrt{\frac{1}{2}}$ BIℓ
9. A non conducting charged ring carrying a charge of q, mass m and radius r is rotated about it's axis, with constant angular speed ω. Find the ratio of magnetic moment with angular momentum.
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$
10. The flux linked with a coil at any instant is given by $\phi B=10t^2-50t+250$. The induced emf at t=3s is
    a) -190V b) -10V c) 10V d) 190V
11. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is a) $\frac{Q}{2}$ b) $\frac{Q}{3}$ c) $\frac{Q}{\sqrt{2}}$ d) Q
12. In a series RLC circuit, the resistance and inductive reactance are same. Then, the phase difference between the voltage and current in the circuit is a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{6}$ d) zero
13. The dimension of $\frac{1}{\mu_o\varepsilon_o}$ is a) a) $[LT^{-1}]$ b) $[L^2T^{-2}]$ c) $[L^{-1}T]$ d) $[L^2T^2]$
14. Fraunhofer lines are an example of _____ spectrum.
    a) line emission b) line absorption c) band emission d) band absorption
15. Which of the following is an electromagnetic wave?
    a) α-rays b) β-rays c) γ-rays d) all of them

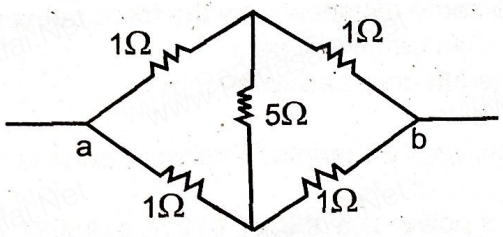
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PART - II

6x2=12

Answer six questions. Q.No.24 is compulsory.

16. Define electrostatic potential at a point.
17. Write any two applications of seebak effect.
18. State ampere circuital law.
19. What is displacement current?
20. Find the effective resistance between points a and b.



21. Mention the ways of producing induced emf.
22. Why does the sky appear blue in colour?
23. Define current sensitivity of galvanometer.
24. A series RLC circuits which resonates at 400 KHz has 80 μH inductor 200 PF capacitor and 50Ω resistor. Calculate Q factor of the circuit.

PART - III

6x3=18

Answer six questions. Q.No.33 is compulsory.

25. Mention the difference between coulomb force and gravitational force.
26. State Kirchoffs current law and voltage law.
27. How do you convert galvanometer into voltmeter?
28. Write any six properties of electromagnetic waves.
29. Give the advantage of AC in long distance power transmission with an example.
30. An electron moving perpendicular to a uniform magnetic field 0.5T undergoes a circular motion of radius 2.50mm. What is the speed of the electron?
31. Write a note on magnetic Lorentz force.
32. Derive an expression for energy stored in a parallel plate capacitor.
33. In a meter bridge experiment, the value of the resistance in the resistance box connected in the right gap is 10Ω. The balancing length is l_1 = 55cm. Find the value of unknown resistance.

PART - IV

5x5=25

Answer all the questions.

34. a) Derive an expression for electrostatic potential due to an electric dipole. **(OR)**
    b) Explain the principle, construction and working of cyclotron.
35. a) Obtain a rotation for the magnetic field at a point along the axis of the circular coil carrying current using Biot-Savart law. **(OR)**
    b) Describe the microscopic model of current and obtain microscopic form of Ohm's law.
36. a) What is emission spectrum? Explain it's type. **(OR)**
    b) Obtain the condition for bridge balance in Wheatstone's bridge.
37. a) Explain the principle, construction and working of a transformer. **(OR)**
    b) Explain the principle, construct and working of Vandegraff generator.
38. a) Explain the Maxwell's modification of Ampere Circuital Law. **(OR)**
    b) (i) Compare the electromagnetic oscillations of Lc Circuit with the mechanical oscillations of Lc circuit.
    (ii) Determine the self inductance of 4000 turn air core solenoid of length 2m and diameter 0.04m.

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